

SIMATS ENGINEERING

ITA05 COMPUTER VISION

LAB ACTIVITY :-

1)Perform basic Image Handling and processing operations on the image.

AIM:- Read an image in python and Convert an Image to Grayscale

PROGRAM:-

```
!pip install opencv-python

import cv2

import matplotlib.pyplot as plt

from google.colab import files

uploaded = files.upload()

image_name = list(uploaded.keys())[0]

img = cv2.imread(image_name)

if img is None:

    print("Error: Image not loaded!")

else:

    rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    cv2.imwrite("gray_image.jpg", gray)

    plt.figure(figsize=(6,6))

    plt.imshow(rgb)

    plt.title("Original Image")

    plt.axis("off")

    plt.show()

    plt.figure(figsize=(6,6))

    plt.imshow(gray, cmap='gray')
```

```
plt.title("Grayscale Image")  
plt.axis("off")  
plt.show()  
files.download("gray_image.jpg")
```

INPUT:-



OUTPUT:-

Grayscale Image



RESULT:-

Thus the program is executed successfully.

2) Perform basic Image Handling and processing operations on the image

AIM:- Read an image in python and Convert an Image to Blur using GaussianBlur.

PROGRAM:-

```
!pip install opencv-python

import cv2

import matplotlib.pyplot as plt

from google.colab import files

uploaded = files.upload()

image_name = list(uploaded.keys())[0]

image = cv2.imread(image_name)

if image is None:

    print("Error: Image not loaded!")

else:

    original = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    blur = cv2.GaussianBlur(image, (5, 5), 10)

    blur_rgb = cv2.cvtColor(blur, cv2.COLOR_BGR2RGB)

    cv2.imwrite("blurred_image.jpg", blur)

    plt.figure(figsize=(6,6))

    plt.imshow(original)

    plt.title("Original Image")

    plt.axis("off")

    plt.show()

    plt.figure(figsize=(6,6))

    plt.imshow(blur_rgb)

    plt.title("Blurred Image")

    plt.axis("off")

    plt.show()
```

```
files.download("blurred_image.jpg")
```

INPUT:-



OUTPUT:-

Blurred Image



RESULT:-

Thus the program is executed successfully.

3) Perform basic Image Handling and processing operations on the image

AIM:-

Read an image in python and Convert an Image to show outline using Canny function.

PROGRAM:-

```
!pip install opencv-python

import cv2

import matplotlib.pyplot as plt

from google.colab import files

uploaded = files.upload()

image_name = list(uploaded.keys())[0]

img = cv2.imread(image_name)

if img is None:

    print("Error: Image not loaded!")

else:

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    edges = cv2.Canny(gray, 100, 200)

    cv2.imwrite("Canny_Edges.jpg", edges)

    plt.figure(figsize=(6,6))

    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))

    plt.title("Original Image")

    plt.axis("off")

    plt.show()

    plt.figure(figsize=(6,6))

    plt.imshow(edges, cmap="gray")

    plt.title("Canny Edge Detection")

    plt.axis("off")

    plt.show()

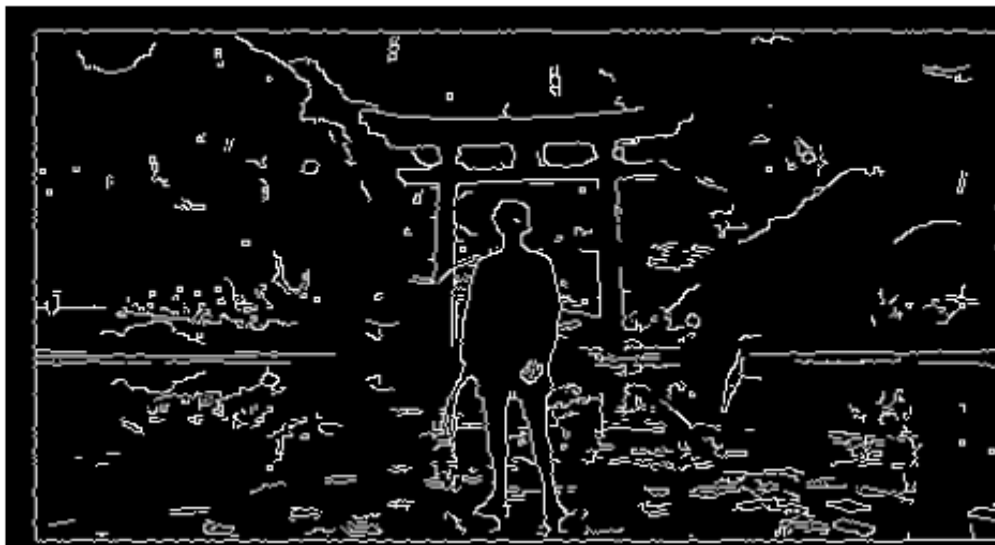
    files.download("Canny_Edges.jpg")
```

INPUT:-



OUTPUT:-

Canny Edge Detection



RESULT:-

Thus the program is executed successfully.

4) Perform basic Image Handling and processing operations on the image

AIM:-

Read an image in python and Dilate an Image using Dilate function.

PROGRAM:-

```
!pip install opencv-python  
  
import cv2  
  
import numpy as np
```

```

import matplotlib.pyplot as plt

from google.colab import files

uploaded = files.upload()

image_name = list(uploaded.keys())[0]

img = cv2.imread(image_name, cv2.IMREAD_GRAYSCALE)

if img is None:

    print("Error: Image not loaded!")

else:

    kernel = np.ones((5,5), np.uint8)

    dilated_img = cv2.dilate(img, kernel, iterations=1)

    cv2.imwrite("Dilated_Image.jpg", dilated_img)

    plt.figure(figsize=(6,6))

    plt.imshow(img, cmap='gray')

    plt.title("Original Image")

    plt.axis("off")

    plt.show()

    plt.figure(figsize=(6,6))

    plt.imshow(dilated_img, cmap='gray')

    plt.title("Dilated Image")

    plt.axis("off")

    plt.show()

    files.download("Dilated_Image.jpg")

```

INPUT:-



OUTPUT:-

Dilated Image



RESULT:-

Thus the program is executed successfully.

5. Perform basic Image Handling and processing operations on the image

AIM:-

Read an image in python and Erode an Image using erode function

PROGRAM:-

```
!pip install opencv-python  
  
import cv2  
  
import numpy as np  
  
import matplotlib.pyplot as plt
```



```
from google.colab import files

uploaded = files.upload()

image_name = list(uploaded.keys())[0]

img = cv2.imread(image_name)

if img is None:

    print("Error: Image not loaded!")

else:

    gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))

    eroded_image = cv2.erode(gray_image, kernel, iterations=1)

    cv2.imwrite("eroded_image.jpg", eroded_image)

    plt.figure(figsize=(6,6))

    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))

    plt.title("Original Image")

    plt.axis("off")

    plt.show()

    plt.figure(figsize=(6,6))

    plt.imshow(eroded_image, cmap="gray")

    plt.title("Eroded Image")

    plt.axis("off")

    plt.show()

    files.download("eroded_image.jpg")
```

INPUT:-



OUTPUT:-

Eroded Image



RESULT:-

Thus the program is executed successfully.